



LarvaLens

UI/UX Design Brief

Visual and Interaction Guide

Public-health visual direction, color and typography tokens, component patterns, mobile layout, accessibility, result hierarchy and prohibited UI patterns.

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Vibe Document 04 - UI/UX Design Brief

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9.1 Design direction

Clean public-health field tool: calm, credible, evidence-first and usable outdoors. Avoid glossy AI gradients, holograms, robots, fake radar effects or decorative dashboards. The interface should make capture, processing status and evidence clearer than branding.

9.2 Visual tokens

Token	Value	Use
Navy	#102A43	Headings, navigation, high-contrast text.
Blue	#1363DF	Primary CTA, links, active controls.
Teal	#0F8B8D	Surveillance/evidence accent and map context.
Green	#208A61	Completed/confirmed state; not "water safe."
Amber	#E59F23	Needs review, medium risk, warnings.
Red	#C94A4A	High risk and blocking errors only.
Background	#F5F8FC	App canvas.
Surface	#FFFFFF	Cards and panels.
Text	#243B53	Body text.

Use Inter or a system sans-serif. Headings are 28/22/18px; body is 16px on mobile and never below 14px for essential information. Use 12px card radius, 8px control radius and subtle borders instead of heavy shadows.

9.3 Key components

- VideoCaptureCard: file/camera input, duration/size status and replacement action.
- LocationConsent: explains why coordinates are requested and supports Skip.
- ModelReadinessBanner: visible only when unavailable/degraded.
- ProcessingTimeline: real stages with current state; no fake percentages.
- ResultHero: probable count, confidence, risk and review status.
- EvidenceFunnel: candidates -> morphology accepted -> temporally supported -> probable larvae.
- TrackEvidenceCard: accepted/rejected badge, detector, larva, non-larva, motion and reason.
- QualityCard: lighting, blur/stability and retake recommendation.
- HotspotMap: clustered/aggregated markers with legend and attribution.
- ReviewDecisionPanel: confirm/reject/inconclusive and notes.

9.4 Mobile layout

- Single column at 360-767px.
- Sticky primary action above the bottom navigation when appropriate.
- Large 44px minimum touch targets.
- Evidence metrics stack vertically; do not compress into unreadable tables.
- Video preview uses 16:9 with contained media.

- Bottom navigation never covers form buttons or browser safe areas.

9.5 Accessibility

- Meet WCAG AA contrast for text and controls.
- Never encode risk or state by color alone; include text and icon.
- All form controls have visible labels and error association.
- Keyboard focus is visible and logical.
- Dialogs trap focus and close with Escape.
- Respect `prefers-reduced-motion`.
- Provide text alternatives for charts and evidence visualizations.
- Use plain language: "No probable larvae detected in this clip" rather than "negative specimen."

9.6 Result screen hierarchy

1. Status and limitations.
2. Probable count, rejected count and confidence.
3. Quality/retake state.
4. Evidence funnel.
5. Track details.
6. Location/map.
7. Model version and audit details in a collapsed technical section.

9.7 Forbidden UI patterns

- Fake live counters, demo charts or seeded hotspots.
- A giant confidence number without what contributed to it.
- "Safe water" or "disease-free" badges.
- Hiding model-unavailable errors behind an infinite spinner.
- Showing precise home coordinates on a public map.
- Exposing private storage object paths or signed URLs in page text.